

*Roberto Festa (festa@units.it)
Department of Philosophy,
Androna Campo Marzio 10, 34123 Trieste, Italy*

Bayesian Inductive Logic, Verisimilitude, and Statistics

Below we will consider the relations between inductive logic and statistics. More specifically, we will show that some concepts and methods of inductive logic may be applied in the rational reconstruction of several statistical notions and procedures and that, in addition, inductive logic suggests some new methods which can be used for different kinds of statistical inference. Although there are several approaches to inductive logic and statistics, here we will focus on some versions of the Bayesian approach and, thereby, on the relations between Bayesian inductive logic and Bayesian statistics. The paper is organized as follows. The subjects of inductive logic and statistics will be shortly illustrated in Section 1, where it will be suggested that statistics can be seen as a special field of inductive logic. Two important theories developed within Bayesian inductive logic are the theory of inductive probabilities, started by Rudolf Carnap in the forties of the past century, and the theory of confirmation: the conceptual relations between such theories and statistics will be considered in Section 2. A recent version of Bayesian inductive logic, proposed by Ilkka Niiniluoto and others, has been developed by using the notion of verisimilitude, introduced in philosophy of science by Karl Popper; the key ideas of the verisimilitudinarian version of Bayesian inductive logic will be illustrated in Section 3, where it will be argued that it provides useful conceptual tools for the analysis of some important kinds of statistical inference.

1. Bayesian inductive logic and Bayesian statistics

1.1. Inductive logic and statistics

Empirical sciences make use of different kinds of inferences leading from a set of one or more statements called premises to another statement, called conclusion. According to a traditional distinction inferences are divided into deductive and inductive. While in deductive inferences the information conveyed by the conclusion is already included in the premises, in inductive inferences the conclusion says something extra with respect to the premises. Hence, even though the truth of the

premises is taken as guaranteed, there is the inevitable risk that the extra information conveyed by the conclusion of an inductive inference may be false. This means that the conclusion of an inductive inference is inevitably uncertain. The systematic analysis of inductive inferences is the subject matter of inductive logic. Suppose that, in an inquiry on the hypothesis H , evidence is expressed by a sentence E , where H cannot be deduced from E . In this case we might ask what are the inductive relations between E and H . A good inductive logic should provide, among other things, a satisfying analysis of the inductive relations between evidence and hypotheses. In particular, it should answer the following questions about the plausibility, confirmation, and acceptability of H in the light of E : (1) What is the degree of *plausibility* of H in the light of E ? (2) What is the degree of *confirmation* conveyed by E to H , i.e., what is the increase of the initial plausibility of H determined by E ? (3) Given evidence E , should we *accept* H , in some suitably specified sense of acceptance? The inductive inferences involved in the questions (1)-(3) may be labelled as plausibilistic, confirmational, and acceptance-based inferences, respectively.

A statistical inference is a particular type of inductive inference where evidence is (the description of) a sample taken from a given population and the hypothesis concerns the whole population or certain samples to be drawn from it. More generally, a statistical inference is an inductive inference where evidence is (the description of) a sequence of trials of a given experimental process and the hypothesis concerns certain parameters of the process or certain future sequences of trials. The main goal of statistics is the formulation of a good theory of statistical inferences, i.e., of a systematic corpus of procedures applicable to any kind of statistical inference.

The above characterization of inductive logic and statistics suggests that statistics is a province of the wide continent of inductive logic. Indeed, if inductive logic is construed as the study of inductive reasoning, wherever and however it occurs, then an ideal inductive logician looks for stable patterns underlying the various forms and areas of inductive reasoning. Hence an ideal inductive logician looks, among other things, for stable patterns underlying the statistical reasoning in empirical sciences. This means that the subject matter of statistics belongs to inductive logic, i.e., that the basic relation between the two disciplines is the very close relation occurring between a whole and a part of it. However, in spite of this intimate relation, starting at least from the beginning of the past century, inductive logic and statistics have undergone a separate development. While inductive logic has been developed within the ground of philosophy, especially within epistemology and philosophy of science, statistics has impetuously grown as a highly specialized discipline, with its own

departments and faculties, in a constant interaction with scientific practice, including the more sophisticated outcomes of probability theory and, more generally, of mathematical research. This separate growth of inductive logic and statistics has led, among other things, to the development of different languages, conceptual systems, and formal tools, so that it is difficult to recognize that in many cases the two disciplines deal with the same problems, and sometimes propose essentially identical solutions. Fortunately, in the last few decades there have been some signals of an opposite tendency, i.e., of an increasing awareness of the close conceptual relations and the potential fruitful interactions between inductive logic and statistics.

1.2. The Bayesian approach to inductive logic and statistics

The most popular approach to inductive logic is the Bayesian approach. The basic idea of *Bayesian inductive logic* (for short: *BIL*) is that any kind of inductive inference should be analysed in terms of inductive – or, equivalently, epistemic – probabilities. This means that any inductive inference from evidence E to the hypothesis H amounts to – or, at least, is based on – the attribution of the epistemic probability $p(H|E)$ to H , where $p(H|E)$ expresses the degree of belief that a rational agent X should attribute to H in the light of E . Within the different versions of BIL this basic idea has been applied in the analysis of plausibilistic, confirmational, and acceptance-based inferences. For instance, the plausibility of H in the light of E is usually identified with $p(H|E)$ and the confirmation conveyed by E to H is usually identified with some measure of the probability increase in the shift from the initial probability $p(H)$ of H to its final probability $p(H|E)$. Different versions of BIL may be distinguished on the basis of the methodological importance attributed to the different kinds of inductive inference. For instance, most supporters of BIL advocate a non-acceptance-based approach, according to which acceptance-based inferences are dispensable both in science and everyday life, where we need only plausibilistic and confirmational inferences. In conflict with this view, some Bayesian inductive logicians defend an acceptance-based approach which emphasizes the methodological indispensability of acceptance-based inferences. Within this approach two basic kinds of acceptance rules have been proposed: (i) purely probabilistic acceptance rules, where the acceptance of a hypothesis H depends *only* on its probability; (ii) decision-based acceptance rules, where the acceptance of H is seen as a cognitive decision depending not only on the probability of H but also on the (suitably defined) cognitive utilities associated with the acceptance of H in the possible states of nature: for instance, the

acceptance rules stated within the so-called cognitive decision theories prescribe to accept the hypothesis which maximizes the expected cognitive utility (Festa 1999b, 2003; Levi, 1967, 1980; Niiniluoto 1987).

While in the last century Bayesianism has been the dominant view among inductive logicians, the orthodoxy among statisticians has been anti-Bayesian. Indeed, the so-called orthodox, or frequentist, statistics developed in the twentieth century – on the track of Jerzy Neyman, Egon S. Pearson, and Ronald A. Fisher (1925, 1935, 1950, 1956) – is based on the explicit rebuttal of the Bayesian idea that statistical inferences can be made by attributing epistemic probabilities to statistical hypotheses. In conflict with this idea, orthodox statisticians maintain that statistical frequencies are the only probabilities to be used in statistical inferences. However, starting from the second half of the twentieth century, the Bayesian approach to statistics has known – thank to the work by Bruno de Finetti (1937/1964; 1974), Harold Jeffreys (1931; 1937), Leonard J. Savage (1954), and many others – an impressive development, so that nowadays *Bayesian statistics* (for short: *BS*) represents a serious alternative to frequentist statistics.

2. Theory of inductive probabilities, confirmation, and statistics

2.1. Inductive methods for multinomial inferences

Most of the work on BIL made by Rudolf Carnap between the beginning of the forties and his death, in 1970, is directed to the development of a theory of epistemic probabilities for the analysis of a well known kind of statistical inference, i.e., multinomial inferences. His theory is usually known under the label of *theory of inductive probabilities* (for short: TIP). Below we will shortly illustrate the notion of inductive method for multinomial inferences – for short: *multinomial method* – and the different strategies used within BS and TIP for the construction of such methods. Afterwards, we will consider some (non-analogical and analogical) multinomial methods introduced within TIP and BS and we will show that, in spite of their different conceptual basis, the methods worked out within TIP are essentially identical to those used within BS.

2.1.1. Multinomial methods within TIP and BS. Consider an experimental process *Ex* whose trials are described by using a qualitative character, or property, $\mathbf{Q} = \{Q_1, \dots, Q_k\}$, where the $k (> 2)$ categories $Q_1, \dots, Q_k$ are mutually exclusive and jointly exhaustive. We say that *Ex* is a *multinomial process* in the case where, for any $Q_i \in \mathbf{Q}$, there is a physical probability $\mathbf{q}_i$ that the outcome of an arbitrary trial of *Ex* is Q_i , where $\mathbf{q}_i$ is

constant across trials – neither influenced by the trial number nor by the outcomes of previous trials. The background knowledge available in many empirical inquiries includes the assumption that the investigated process Ex is a multinomial process. However, typically a researcher does not know the true value of the parameter vector $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_k)$ of Ex , but can only formulate different kinds of hypotheses about $\mathbf{q}$ and the results of future trials of Ex . The inductive inferences concerning such hypotheses will be called *multinomial inferences*. More precisely, a multinomial inference about Ex amounts to the determination of the epistemic probability $p(H|E)$, where each of the sentences H and E may concern either the value of $\mathbf{q}$ or the result of some sequence of trials of Ex . A *multinomial method* is an inductive procedure allowing to make any kind of multinomial inference about any multinomial process Ex .

Within BS the starting point for the definition of a multinomial method is given by a suitable representation of the researcher's initial opinions about the value of the parameter vector $\mathbf{q}$ of the investigated multinomial process Ex , i.e., by an appropriate *prior distribution* $F(q)$ of epistemic probabilities on the possible values of $\mathbf{q}$. A different strategy is adopted within TIP, where the starting point for the construction of a multinomial method is given by the definition of an *inductive rule* R specifying, for any possible sequence e_n of outcomes of n trials of Ex and any category $Q_i \in \mathbf{Q}$, the *predictive probability* $p(Q_i|e_n)$ of the hypothesis that the result of the next trial of Ex will belong to Q_i . Then the predictive probabilities $p(Q_i|e_n)$ may be used to determine the prior probability $p(e_n)$ of any sequence e_n of outcomes of n future trials of Ex , the posterior probability $p(e_n|e_m)$, where e_m describes a sequence of outcomes of m past trials of Ex , and any other epistemic probability $p(H|E)$ associated to Ex . In Carnap's terminology predictive probabilities $p(Q_i|e_n)$ are referred to as the *special values* of a multinomial method, where "special" refers to their special importance in the definition of a multinomial method.

The conceptual differences between the strategies for the construction of multinomial methods adopted within BS and TIP can be construed as the differences between a *globalistic* and a *predictivistic* approach to multinomial inferences. Indeed, the strategy adopted within BS is based on the attribution of appropriate epistemic probabilities to the possible values of the parameter vector $\mathbf{q}$ of the multinomial process Ex which is, as it were, a *global* feature of Ex . On the contrary, the strategy adopted within TIP is based on the attribution of appropriate epistemic probabilities to any possible *prediction* concerning a future trial of Ex in the light of any possible sequence of past trials of Ex .

Although this conceptual difference between TIP and BS cannot be neglected, it should be recalled that there is also an important conceptual equivalency between TIP and BS. We refer to the circumstance that an

important class of multinomial methods investigated within TIP, i.e., the exchangeable multinomial methods, is essentially identical to the multinomial methods considered within BS. Let n_i denote the frequency of Q_i in sequence e_n of trials of Ex , i.e., the number of outcomes in e_n belonging to the category Q_i . Then a multinomial method for Ex is called *exchangeable* in the case where, for any n and any e_n , the prior probability $p(e_n)$ depends *only* on $n_1, \dots, n_k$ – and not on the order in which the outcomes $Q_1, \dots, Q_k$ occur in e_n . De Finetti's representation theorem implies that any exchangeable multinomial method I for Ex is essentially identical to a corresponding (unique) prior distribution $F_I(q)$ defined on the parameter vector $\mathbf{q}$ of Ex (de Finetti 1937/1964, 1974). This means that the exchangeable multinomial methods worked out within TIP are essentially identical to the multinomial methods considered within BS.

2.1.2. *The equivalence between GC-methods and Dirichlet distributions.*

A well known family of exchangeable multinomial methods is given by *generalized Carnapian methods* (for short: *GC-methods*), which are characterized by the following inductive rule (Carnap and Stegmüller 1959):

$$(GC\text{-rule}) \quad p(Q_i | e_n) = \frac{n_i + \lambda \gamma_i}{n + \lambda}$$

where $\gamma_i > 0$, $\sum \gamma_i = 1$, and $0 \leq \lambda \leq \infty$.

The multinomial methods used within BS are typically based on the so-called Dirichlet distributions, which can be defined on the parameter vector $\mathbf{q}$ of any multinomial process Ex . It can be easily proved that – as a consequence of de Finetti's theorem – any GC-method is equivalent to a corresponding Dirichlet distribution. It may be interesting to point out that, in spite of its relative triviality, the equivalence between GC-methods and Dirichlet distributions is not mentioned in Carnap's papers and, more generally, that it passed unnoticed until 1965 (Good 1965; Festa 1993). This circumstance clearly reveals the almost surprising lack of interactions between TIP and BS in the forties and fifties of the past century. However, starting from the seventies, the conceptual relations between TIP and BS have been increasingly understood.

2.1.3. *The axiomatization of GC-methods and Dirichlet distributions.*

Between the fifties and the seventies of the past century, inductive

logicians and Bayesian statisticians succeeded in axiomatizing GC-methods and Dirichlet distributions.

Axiomatizing a class of inductive methods amounts to proving that the class includes all and only those methods which satisfy a given set of axioms. In particular, GC-systems have been axiomatized by several authors (Carnap and Stegmüller 1959; Carnap 1980, § 19; Kuipers 1978, Ch. 5). Their results may be summarized as follows:

(T.1) An exchangeable multinomial method I for a family $\mathbf{Q} = \{Q_1, \dots, Q_k\}$ of $k > 2$, is a GC-method in the case where the predictive probabilities $p(Q_i|e_n)$ determined on the basis of I satisfy the following principles:

(IP) *Principle of initial possibility.* $p(Q_i) \equiv \gamma_i > 0$.

(RR) *Principle of restricted relevance.* If e_n and e'_n are such that $n_i = n'_i$, then $p(Q_i|e_n) = p(Q_i|e'_n)$

Recalling that Dirichlet distributions are prior distributions defined on the parameter vector $\mathbf{q}$ of a multinomial process Ex , one may ask whether Dirichlet distributions can be axiomatized on the basis of appropriate principles expressing plausible constraints on the way in which we assign our prior probabilities to the possible values of $\mathbf{q}$. We will see below that a positive answer to this question can be given.

Axiomatizing a class of prior distributions amounts to proving that the class includes all and only those distributions which satisfy a given set of axioms. In particular, Dirichlet distributions have been axiomatized by Fabius (1973). His axiomatization is based on the concept of *neutrality*. Put simply, given the proportions $\mathbf{q}_1, \dots, \mathbf{q}_k$ – which are defined as non-negative parameters satisfying the constraint $\sum \mathbf{q}_i = 1$ – the proportion $\mathbf{q}_1$ is said to be *neutral* w.r.t. $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_k)$ when no information on the value of $\mathbf{q}_1$ can influence our beliefs concerning “the manner in which the remaining proportions $\mathbf{q}_2, \dots, \mathbf{q}_k$ *proportionally* divide the remainder of the unit interval” (Connor and Mosimann, 1969, p. 196). Fabius’s axiomatization of Dirichlet distributions (*ibid.*) can be stated as follows:

(T.2) A prior distribution $F(\mathbf{q})$ on the parameter vector $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_k)$ of a multinomial process Ex is a Dirichlet distribution in the case where $F(\mathbf{q})$ satisfies the following principle:

(N) *Principle of neutrality.* $F(\mathbf{q})$ is such that any $\mathbf{q}_i$ is *neutral* w.r.t. $\mathbf{q}$.

An important difference between the axiomatizations of GC-methods and Dirichlet distributions should be noticed. The principles (IP) and (RR), used in the axiomatization of GC-methods, are *predictive* principles concerning the behaviour of the inductive rule used to determine the predictive probabilities $p(Q_i|e_n)$; on the contrary, the principle (N), used in the axiomatization of Dirichlet distributions, is a *global* principle concerning our opinions about the global feature $\mathbf{q}$ of Ex , i.e., the probabilities attributed to the possible values of $\mathbf{q}$.

Given the essential identity between GC-methods and Dirichlet distributions, we might say that each of the theorems (T.1) and (T.2) provides an axiomatic foundation both of GC-methods and Dirichlet distributions. However, while the axiomatization stated in (T.1) provides a *predictivistic justification* of GC-methods and Dirichlet distributions, the axiomatization stated in (T.2) provides a *globalistic justification* of such methods.¹

2.1.4. The non-analogical nature of GC-methods and Dirichlet distributions. The principles (RR) and (N), used in the axiomatizations of GC-methods and Dirichlet distributions immediately reveal the non-analogical nature of such multinomial methods. Indeed, the principle (RR), used in the axiomatization of GC-methods, implies that the predictive probabilities $p(Q_i|e_n)$ of a GC-method depend on the empirical evidence e_n only via n and n_i . In other words, (RR) implies that $p(Q_i|e_n)$ should in no way be affected by the empirical frequencies n_j – with $j \neq i$ – in e_n ; this independently of the possible, more or less strong, similarity occurring between categories Q_j and Q_i . This means that GC-methods cannot take into account any analogy by similarity, i.e., that GC-methods are not analogical methods. Similar anti-analogical intuitions seem to underlie the principle (N), used in the axiomatization of Dirichlet distributions. Indeed (N) excludes any analogy by similarity by requiring that our beliefs about $\mathbf{q}_i$ should in no way be affected by information concerning the value of the fraction $\mathbf{q}_j/(1 - \mathbf{q}_i)$ – with $j \neq i$ – in the remainder $(1 - \mathbf{q}_i)$ of the unit interval; this independently of the possible, more or less strong, similarity occurring between categories Q_j and Q_i .

2.1.5. Analogical methods for one property. Although most of Carnap's work on TIP was devoted to the investigation of GC-methods and other

1

Notes

. On the axiomatic foundation of GC-methods and Dirichlet distribution – and the justification of the use of such multinomial methods in empirical inquiries – see Festa (1993, Ch. 6 and 7.6).

non-analogical multinomial methods, the construction of analogical inductive methods applicable to *multicategorical experiments* – i.e., to experiments whose outcomes are described by one or more categories – has always been on Carnap's agenda.

Let us consider, first of all, the problem of defining appropriate *analogical methods for one property*. Such methods may be introduced w.r.t. to any kind of experimental process described by a property $\mathbf{Q} = \{Q_1, \dots, Q_k\}$ with $k > 2$. This means that *Ex* might be a multinomial process, but it might also be a different kind of physical process, or a process whose nature is not exactly known.

Given three categories Q_i , Q_j , and Q_k belonging to $\mathbf{Q}$, suppose that the similarity between Q_j and Q_i is higher than that between Q_k and Q_i . If our predictive probabilities are determined by using a GC-method, then equality $p(Q_i|Q_j) = p(Q_i|Q_k)$ holds; more generally, for any sequence e_n of trials of *Ex*, equality $p(Q_i|e_n Q_j) = p(Q_i|e_n Q_k)$ holds. This means that, if a GC-method is used, then the predictive probabilities are not affected by the similarity relations occurring among Q_i , Q_j , and Q_k . However, in certain cases we might require that our inductive method is sensible to such similarity relations; for instance, we might require that it satisfies the following *weak principle of analogy by similarity*:

(WAS) For any triple Q_i , Q_j , and Q_k such that Q_j is more similar than Q_k to Q_i , $p(Q_i|Q_j) > p(Q_i|Q_k)$.

In some cases we might also require that our inductive method satisfies the following *strong principle of analogy by similarity*:

(SAS) For any sequence e_n of trials of *Ex* and any triple Q_i , Q_j , and Q_k such that Q_j is more similar than Q_k to Q_i , $p(Q_i|e_n Q_j) > p(Q_i|e_n Q_k)$.

We will say that an inductive method is a *weakly analogical method* in the case where it satisfies (WAS) or akin weak principles of analogy by similarity, while it is a *strongly analogical method* in the case where it satisfies (SAS) or akin strong principles of analogy by similarity. Within TIP the problem of defining appropriate analogical methods for one property has been investigated by Carnap (1963, 1980), Niiniluoto (1981), Costantini (1983), Kuipers (1984a, b; 1988), Skyrms (1993a), Festa (1996), and others. The methods proposed by the just mentioned authors can be classified as follows:

(1) **STRONGLY ANALOGICAL NON-EXCHANGEABLE METHODS.** The analogical methods proposed by Carnap (1980), Niiniluoto (1981), Costantini (1983), and Kuipers (1984a, b; 1988) are of this kind. Their methods are

introduced in the TIP-style, i.e., on the basis of simple and intuitively transparent inductive rules specifying the predictive probabilities $p(Q_i|e_n)$. Such inductive rules are explicitly designed to guarantee the satisfaction of some strong principles of analogy by similarity, such as (SAS) (Carnap), or akin principles (Niiniluoto and Kuipers). Unfortunately, the above methods exhibit also three related and not very attractive features: (i) they are not exchangeable; (ii) thereby, they are not multinomial methods; indeed, the adoption of a non-exchangeable method implies that the investigated process Ex is not multinomial, since any prior distribution $F(q)$ on the parameter vector q of a multinomial process is essentially identical to an exchangeable method; (iii) finally, it is not clear whether the adoption of such methods imposes some constraint on the (non-multinomial) physical structure of Ex .

(2) WEAKLY ANALOGICAL EXCHANGEABLE METHODS. A method of this kind has been proposed by Skyrms (1993a). Such method – which will be called Sk – is introduced in the BS-style, i.e., as a prior distribution $F_{Sk}(q)$ on the parameter vector q of a particular multinomial process Ex , given by a wheel of fortune whose possible outcomes are described by the property $Q = \{\text{North, East, South, West}\}$. More precisely, $F_{Sk}(q)$ is given by a mixture of four Dirichlet distributions defined on q . This implies that Sk is exchangeable, i.e., that Sk is a multinomial method. One can easily check that Sk is weakly analogical since it satisfies (WAS). However, none of the strong principles of analogy so far proposed in the literature is satisfied by Sk (Festa 1996). Finally, it is not clear whether Sk can be generalized so to obtain a class of strongly analogical methods applicable to different kinds of multinomial process.

(3) STRONGLY ANALOGICAL EXCHANGEABLE METHODS. A method of this kind has been proposed by Festa (1996). Also Festa's method – which will be called Fe – is introduced in the BS-style, i.e., as a prior distribution $F_{Fe}(q)$ on the parameter vector q of a particular multinomial process Ex , described by the property $Q = \{Q_1, Q_2, Q_3\}$ with three categories. More precisely, $F_{Fe}(q)$ is given by a mixture of two Dirichlet distributions defined on q . This implies that Fe is exchangeable, i.e., that it is a multinomial method. Moreover, it can be proved that Fe is strongly analogical since it satisfies (SAS). However, together with such attractive features, Fe shares with Sk an unpleasant feature; indeed, it is not clear whether and how Fe can be generalized to different kinds of multinomial process.

The research on analogical methods for one property seems to be an important field for the cooperative development of TIP and BS: indeed, some of the above mentioned analogical methods have been developed

by using the conceptual tools both of TIP and BS. However, one has to recognize that so far the research on analogical methods for one property has not led to outcomes comparable with those of the research on non-analogical methods and, in particular, on GC-methods and Dirichlet distributions. Among other things, it should be recalled that, so far, (i) no general class of exchangeable analogical methods has been proposed, and (ii) no axiomatic foundation of exchangeable analogical methods has been provided.

2.1.6. Analogical methods for many properties. Another interesting problem considered within TIP is the definition of appropriate *analogical methods for many properties*. Such methods have been investigated with reference to multinomial processes whose possible outcomes are cross classified on the basis of two or more properties (Carnap 1963; Hesse 1964; Maher 2000, 2001; Romeyn 2005, 2006). In the most simple case, the outcomes are described by using two properties $\mathbf{F} = \{F_1, F_2\}$ and $\mathbf{G} = \{G_1, G_2\}$ where $F_2 \equiv \text{non-}F_1$ and $G_2 \equiv \text{non-}G_1$. Given two trials a and b of a multicategorical experiment Ex , we might require that our predictive probabilities satisfy the following analogical principle:

$$(A) \quad p(G_1b|F_2G_1a \ \& \ F_1b) > p(G_1b|F_2G_2a \ \& \ F_1b)$$

Suppose, for instance, that Ex is given by the random drawing of an individual from the population of swans and that the members of such population are described by the properties $\mathbf{F}$ and $\mathbf{G}$, where F_1 stands for “Australian” and G_1 for “white”. Then a and b will refer to individuals randomly drawn from the population of swans. With reference to this example, the intuitive meaning of (A) can be expressed as follows: the probability that the next observed Australian swan is white, given that a previously observed non-Australian swan is white, is greater than the probability of such hypothesis given that a previously observed non-Australian swan is non-white.

If an inductive method satisfies (A) or other akin analogical principles, we will say that it is an analogical method for many properties. Several analogical methods for many properties have been introduced in the literature. While the investigation of analogical methods for many properties was initially carried out using only the conceptual tools of TIP (Carnap 1963; Hesse 1964; Niiniluoto 1980, 1988), the research on this subject made in the last decade is characterized by the extensive use also of the conceptual tools of BS (Maher 2000, 2001; Romeyn 2005, 2006). In particular, such research has shown that analogical methods for many properties have close – and still not fully explored – relations with the

statistical research on contingency tables (Albert and Gupta 1983; Epstein and Feinberg 1992; Good 1965, 1983; Lindley 1964).²

2.2. Inductive methods for non-multinomial statistical inferences

While the research on analogical methods has not led to outcomes comparable with those of the research on GC-methods and Dirichlet distributions, a much more successful story can be told about the research made on other two issues which were on Carnap's agenda. We are referring to the construction of inductive methods for a value continuum and of inductive methods sensitive to analogy by proximity. Although here there is no room for this successful story, the reader can find an lucid survey of the research in this area in Skyrms (1996).³ Among other things, Skyrms clearly shows that such research is characterized by a deep integration of TIP and BS methods. For instance, within three years of Carnap's death, the problem of the construction of inductive methods for a value continuum was "completely solved in a way quite consonant with Carnapian techniques" (Skyrms ib., p. 321), by three Bayesian statisticians, i.e., Ferguson (1973) and Blackwell and MacQueen (1973).

2.3. Confirmation and statistics

As said in Section 1.2, the confirmation conveyed by evidence E to a hypothesis H is usually identified with some measure of the probability increase in the shift from the initial probability $p(H)$ of H to its final probability $p(H|E)$. Two popular incremental measures of confirmation are the probability difference $c_d(H, E) \equiv p(H|E) - p(H)$ and the probability ratio $c_r(H, E) \equiv p(H|E)/p(H)$. Another popular incremental measure is defined in terms of the initial and final odds of H , i.e., in terms of $o(H) \equiv p(H)/(1 - p(H))$ and $o(H|E) \equiv p(H|E)/(1 - p(H|E))$. We are referring to the so-called *odds ratio* $c_{or}(H, E) \equiv o(H|E)/o(H)$. An attractive feature of $c_{or}(H, E)$ is given by the easily proved equality $c_{or}(H, E) = p(E|H)/p(E|\neg H)$. The quantity $p(E|H)/p(E|\neg H)$ is commonly known as *Bayes factor* (in favour of H). It should be noticed that, while $c_{or}(H, E)$ is well defined only in the case where

². For an explicit reference to the relation between Carnap's analogical methods for two properties and the problem of estimating cell probabilities in the case of a two-dimensional contingency table, see Costantini and Garibaldi (1996).

³. See also Skyrms (1991, 1993b).

$p(H)$ is positive, in principle the Bayes factor $p(E|H)/p(E|\neg H)$ may be well defined also in the case where $p(H)$ is zero. More generally, one immediately sees that the initial probability of H is not necessary for the determination of the Bayes factor in favour of H . These feature of Bayes factor explains why it has been considered as an attractive measure of confirmation for statistical hypotheses, also by a number of non-Bayesian statisticians, who would be scarcely inclined to adopt confirmation measures stated in terms of the initial or final probabilities of statistical hypotheses.

Although the research on the confirmation of scientific hypotheses has been carried out mainly within the general framework of inductive logic (Festa 1999a, 2009; Festa et al. 2009; Fitelson 1999), in the last few decades the concept of confirmation – or, equivalently, empirical support – has attracted an increasing attention among statisticians. For instance, several frequentist statisticians have suggested that the so-called p -values of statistical hypotheses can be construed as an appropriate measure of their degree of empirical support. The first Bayesian statistician who has devoted a lot of attention to the confirmation measures suggested within inductive logic – and to the possibility to apply such measures to statistical hypothesis – is I. J. Good. Among other things, Good provides a thorough analysis of Bayes factor and suggests a Bayesian rational reconstruction of the measures of corroboration proposed by Karl Popper as an alternative to Bayesian measures of confirmation.⁴

3. Verisimilitude and statistics

3.1. Verisimilitude and inductive logic: towards a verisimilitudinarian Bayesian inductive logic

A recent version of BIL, which may be called *verisimilitudinarian BIL*, is based on the integration, within the conceptual framework of BIL, of the notion of verisimilitude. The first formal notion of verisimilitude – or approximation to the truth – was worked out by Karl Popper within his falsificationist and anti-inductivist philosophy of science (Popper 1963). Afterward such notion, and its methodological applications, have been extensively investigated within the post-Popperian theories of

⁴. See especially Good (1950, 1960, 1960/1968, 1961, 1968, 1975, 1980, 1981a, b, 1982, 1983, 1984, 1985a, b, 1989a, b, c, 1991, 1997) and Good and McMichael (1984).

verisimilitude, emerged since 1975.⁵ Two basic issues dealt with by such theories are the following:

(1) The analysis of the logical problem of verisimilitude, in order to provide appropriate notions of verisimilitude and distance from the truth. Such notions should allow us to compare any two theories w.r.t. their closeness to the truth, even in the case where they are false. Post-Popperian theories of verisimilitude generally admit that adequate notions of verisimilitude and distance from the truth should leave room for the possibility that, under certain conditions, a false theory is closer to the truth than a true one.

(2) The analysis of the epistemic problem of verisimilitude, in order to define the notions of estimated verisimilitude and estimated distance from the truth. Such notions should allow to compare, on the basis of the available data, any two theories w.r.t. their estimated closeness to the truth, even in the case where they have been falsified by such data. It is widely agreed that adequate notions of estimated verisimilitude and estimated distance from the truth should allow that, under certain conditions, the estimated verisimilitude of a falsified theory is higher than that of a non-falsified one.

The notion of verisimilitude may be used in the analysis of all the three kinds of inductive inference defined in Section 1.2, i.e., plausibilistic, confirmational, and acceptance-based inferences. In fact one may define the plausibility of H in the light of E as the estimated verisimilitude of H in the light of E and may identify the degree of confirmation conveyed by E to H with an appropriate measure of the increase of the initial estimated verisimilitude of H determined by E . Finally, one may state verisimilitudinarian rules of inductive acceptance based on the idea that verisimilitude is the proper cognitive utility of science, so that, within a set of rival hypotheses, one should accept the hypothesis which maximizes the expected verisimilitude.

3.2. Verisimilitudinarian Bayesian inductive logic and statistics

In the last twenty years the notions and principles of verisimilitudinarian BIL have been applied in the rational reconstruction of a number of widely

⁵. An excellent survey of the post-Popperian theories of verisimilitude is provided by Niiniluoto (1998).

used statistical procedures. Some of these applications are shortly illustrated below.

(1) THE CHOICE OF THE OPTIMUM MULTINOMIAL METHODS. Multinomial inferences might be defined as statistical inferences aiming at establishing the true value of the parameter vector $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_k)$ of a multinomial process *Ex.* A major problem for the Bayesian analysis of multinomial inferences – which has been thoroughly investigated both in TIP and BS – is the choice of an optimum prior distribution $F^*(q)$ on $\mathbf{q}$. Festa (1993, 1995) considers this problem with reference to Dirichlet distributions (or, equivalently, GC-methods) and suggests a verisimilitude solution, inspired by the intuitive idea that the optimum Dirichlet distribution $F^*(q)$ should be chosen so that the estimates of $\mathbf{q}_1, \dots, \mathbf{q}_k$ based on $F^*(q)$ approach the truth most effectively. This means that a good reason for selecting $F^*(q)$, within the class of Dirichlet distributions, is that there are grounds to believe that $F^*(q)$ is the optimum tool for achieving a high degree of verisimilitude. The main points of Festa's verisimilitude solution can be informally stated as follows: (i) the efficiency of a Dirichlet distribution (GC-method) in approaching the truth depends on the degree of order in the sector of the universe under investigation; (ii) hence, the adoption of a given Dirichlet distribution (GC-method) as the optimum multinomial method – i.e., as the presumably most efficient tool to approach the truth – should be based on appropriate presuppositions about the degree of order in the investigated sector of the universe; (iii) such presuppositions are given by the scientists' initial estimates – grounded on their background knowledge – of the degree of order of the sector of the universe under examination within a given empirical inquiry.

(2) BAYESIAN POINT AND INTERVAL ESTIMATES. Given a distribution of epistemic probabilities on the relevant set of statements, one can calculate, on the basis of available evidence *E*, the expected verisimilitude and the expected distance from the truth of any hypothesis *H* under inquiry. One can construct a theory of inductive acceptance based on the idea that the distance from the truth is the cognitive loss whose expected value should be minimized when a hypothesis is selected for inductive acceptance. This idea has been applied to the rational reconstruction of certain standard results about point estimation obtained in Bayesian statistics. More specifically, Niiniluoto (1982a, 1987, pp. 27-29) shows that, given a suitable loss function, accepting a Bayesian point estimate is equivalent to minimizing the expected distance from the truth. Furthermore, Niiniluoto (1982b, 1986, 1987, pp. 430-41) and others (Festa 1986; Maher 1993, pp. 143-47) extend this perspective to Bayesian interval estimation by approaching interval estimation in decision-theoretic terms, where the loss

of accepting an interval is defined by the distance of this interval from the truth.

(3) BAYESIAN ANALYSIS OF OBSERVATIONAL ERRORS. The expected verisimilitude of a hypothesis H can be seen as a reasonable estimate of the degree of verisimilitude of H , while the probable verisimilitude of H expresses our degree of belief in the possibility that H is highly verisimilar, i.e., that the verisimilitude of H exceeds a fixed threshold (Niiniluoto 1987, Ch. 7.3). The notions of expected and probable verisimilitude have been applied to the analysis of significant statistical problems such as the problem of observational errors. This problem typically arises in the cases where the evidence is obtained by measuring a given quantity: in fact, in such cases there is a positive probability that, due to errors in measurement, the result of measurement deviates from the true value of the quantity. After discussing in general terms the possibility of appraising verisimilitude on the basis of false evidence, Niiniluoto (1987, Ch. 7.4) suggests that the classical Bayesian treatment of observational errors can be understood as an attempt to evaluate the expected and probable verisimilitude of hypotheses on the basis of evidence which is known (with probability one) to be erroneous.

(4) PROBLEMS OF CURVE-FITTING AND REGRESSION ANALYSIS. In many cognitive problems – for instance, whenever the inquiry is based upon counterfactual idealizing conditions – all the relevant hypotheses under inquiry are known to be false. A well known example of this sort of cognitive problems is given by the statistical problems of curve-fitting, where all the relevant quantitative hypotheses, expressed by sufficiently simple linear functions, are known to be false. A verisimilitude interpretation of the typical statistical methods based on the Least Square Difference between a curve and a finite set of data points is provided by Niiniluoto (1987, Ch. 7.5). He suggests also that verisimilitude could be used for the rational reconstruction of the statistical methods of regression analysis which are applied to problems in which all the relevant hypotheses are false (idealizations) and the evidence is known to be erroneous (Niiniluoto 1987, p. 289).

(5) APPROACHING THE STATISTICAL TRUTH BY QUALITATIVE THEORIES. In the last three decades several measures of verisimilitude have been defined for different kinds of qualitative and quantitative theories, and their methodological applications have been thoroughly explored. Above we have seen, with reference to quantitative hypotheses, that the notion of verisimilitude can be fruitfully applied in the analysis of some important kinds of statistical inferences. However, recently it has been argued that

also qualitative theories may have interesting statistical applications (Festa 2007a, b, 2008). More precisely, it has been argued that qualitative theories stated in qualitative languages with two or more properties can be used in the description of the statistical structure of cross classified populations and that their adequacy in this task can be evaluated by appropriate measures of their *statistical verisimilitude*. Among other things, Festa's notion of statistical verisimilitude has been used for the rational reconstruction of some aspects of the so-called prediction logic, a relatively recent approach to the statistical analysis of cross classified populations developed by the statisticians and social scientists David K. Hildebrand, James D. Laing, and Howard Rosenthal in a series of papers published between 1974 and 1977, which culminated in their not yet adequately appreciated book *Prediction Analysis of Cross Classification* (1976).

The above, necessarily incomplete, survey of some research areas on the border between inductive logic and statistics shows the abundance of conceptual relations between the two disciplines. It also suggests that inductive logicians and statisticians should engage in a serious effort to reduce the inessential differences in their languages and conceptual systems and to explore the numerous possibilities of interaction between – and cooperative development of – the two disciplines.⁶

Acknowledgements. I like to thank an anonymous referee for helpful comments on the first draft of this paper.

References

Albert, J. H. and A. K. Gupta (1983). Bayesian Estimation Models for 2 H 2 Contingency Tables Using Mixtures of Dirichlet Distributions. *Journal of the American Statistical Association* 78, 708-717

6

. For an excellent survey of the conceptual relations between statistics and inductive logic – and, more generally, between statistics and philosophy of science – see Good (1988).

- Blackwell D. and MacQueen J. B. (1973). Ferguson Distributions via Polya Urn Schemes. *Annals of Statistics* 1, 353-355
- Carnap, R. (1963). Variety, Analogy and Periodicity in Inductive Logic. *Philosophy of Science* 30, 222-227
- Carnap, R. (1980). 'A Basic System of Inductive Logic, Part II. In R. Jeffrey (ed.), *Studies in Inductive Logic and Probability*, vol. II. Berkeley: University of California Press, pp. 7-155
- Carnap, R. and Stegmüller, W. (1959). *Induktive Logik und Wahrscheinlichkeit*. Wien: Springer-Verlag
- Connor, R. J. and Mosimann, J. E. (1969). Concepts of Independence for Proportions with a Generalization of the Dirichlet Distribution. *JASA* 64, 194-206
- Costantini, D. (1983). Analogy by Similarity. *Erkenntnis* 20, 103-114
- Costantini, D. and Garibaldi, U. (1996). Predictive Laws of Association in Statistics and Physics. *Erkenntnis* 45, 259-261
- de Finetti, B. (1937/1964). La prévision: ses lois logiques, ses sources subjectives. *Annales Institute Henri Poincaré* 7, 1-68. English translation in Kyburg H. E. Jr. and Smokler, H. (eds.), *Studies in Subjective Probability*. New York: Wiley, pp. 93-158
- de Finetti, B. (1974). *Probability, Induction and Statistics*. New York: Wiley
- Epstein, L. D. and Feinberg S. E. (1992). Bayesian Estimation in Multidimensional Contingency Tables. In P. K. Goel and N. S. Iyengar (eds.), *Bayesian Analysis in Statistics and Econometrics*. New York: Springer, pp. 27-41
- Fabius, J. (1973). Two Characterizations of the Dirichlet distributions. *Annals of Mathematical Statistics* 1, 583-587
- Ferguson, T. (1973). A Bayesian Analysis of Some Non-Parametric Problems. *Annals of Statistics* 1, 209-230
- Festa, R. (1986). A Measure for the Distance Between an Interval Hypothesis and the Truth. *Synthese* 67, 273-320
- Festa R. (1993). *Optimum Inductive Methods. A Study in Inductive Probability, Bayesian Statistics, and Verisimilitude*. Dordrecht: Kluwer
- Festa R. (1995). Verisimilitude, Disorder, and Optimum Prior Probabilities. In T. K. Kuipers and A. R. Mackor (eds.), *Cognitive Patterns in Science and Common Sense*. Rodopi: Amsterdam, pp. 299-320
- Festa R. (1996). Analogy and Exchangeability in Predictive Inferences. *Erkenntnis* 45, 229-252
- Festa R. (1999a). Bayesian Confirmation. In M. C. Galavotti e A. Pagnini (eds.), *Experience, Reality, and Scientific Explanation*. Dordrecht: Kluwer, pp. 55-87
- Festa R. (1999b). Scientific Values, Probability, and Acceptance. In R. Rossini Favretti, G. Sandri e R. Scazzieri (eds.), *Incommensurability and Translation*. Cheltenham: Edward Elgar, pp. 323-338
- Festa R. (2003). Induction, Probability, and Bayesian Epistemology. In L. Haaparanta and I. Niiniluoto (eds.), *Analytic Philosophy in Finland*. Amsterdam: Rodopi, pp. 251-284

- Festa R. (2007a). Verisimilitude, Cross Classification, and Prediction Logic. Approaching the Statistical Truth by Falsified Qualitative Theories. *Mind and Society* 6, 37-62
- Festa R. (2007b). The Qualitative and Statistical Verisimilitude of Qualitative Theories. *La Nuova Critica* 47-48, 91-114
- Festa R. (2008). Verisimilitude, Qualitative Theories, and Statistical Inferences. In M. Sintonen, S. Pihlström and P. Raatikainen (eds.), *Approaching Truth: Essays in Honour of Ilkka Niiniluoto*. London: College Publications, pp. 143–177
- Festa R. (2009). “For Unto Every One That Hath Shall Be Given”: Matthew Properties for Incremental Confirmation’, *Synthese*. Forthcoming
- Festa R., V. Crupi, and C. Buttasi (2009). ‘Towards a Grammar of Bayesian Confirmation’. In M. Dorato, M. R. Adei, and M. Suárez (eds.), *EPSA Epistemology and Methodology of Science*. Berlin: Springer, pp. 120-155.
- Fisher R. A. (1925). *Statistical Methods for Research Workers*. Edinburgh: Oliver & Boyd
- Fisher R. A. (1935). *The Design of Experiments*. Edinburgh: Oliver & Boyd
- Fisher R. A. (1950). *Contributions to Mathematical Statistics*. New York: Wiley
- Fisher R. A. (1956). *Statistical Methods and Scientific Inference*. Edinburgh: Oliver & Boyd
- Fitelson, B. (1999). The Plurality of Bayesian Measures of Confirmation and the Problem of Measure Sensitivity. *Philosophy of Science* 66, S362-S378
- Good, I. J. (1950). *Probability and the Weighing of Evidence*. London: Charles Griffin and Co.
- Good, I. J. (1960). The Paradox of Confirmation. *BJPS* 11, 145–149
- Good, I. J. (1960/1968). Weight Of Evidence, Corroboration, Explanatory Power, Information, and the Utility Of Experiments. *JRSS B* 22, 319-331; Corrigenda 30, 203
- Good, I. J. (1961). The Paradox of Confirmation (II). *BJPS* 12, 63–64
- Good, I. J. (1965). *The Estimation of Probabilities*. Cambridge, Mass.: The MIT Press
- Good, I. J. (1968). Corroboration, Explanation, Evolving Probability, Simplicity, and A Sharpened Razor. *BJPS* 19, 123-143
- Good, I. J. (1975). Explicativity, Corroboration, and the Relative Odds of Hypotheses. *Synthese* 30, 39-73 and 83-93
- Good, I. J. (1980). Another Relationship Between Weight of Evidence and Errors of the First and Second Kinds. *JSCS* 10, 315-316
- Good, I. J. (1981a). The Weight of Evidence Provided by Uncertain Testimony or from an Uncertain Event. *JSCS* 13, 56-60
- Good, I. J. (1981b). An Error by Peirce Concerning Weight of Evidence. *JSCS* 13, 155-157
- Good, I. J. (1982). A Good Explanation of an Event Is Not Necessarily Corroborated by the Event. *Philosophy of Science* 49, 251-253
- Good, I. J. (1983). *Good Thinking*. Minneapolis: The University of Minnesota Press
- Good, I. J. (1984). The Best Explication for Weight of Evidence. *JSCS* 19, 294-299

- Good, I. J. (1985a). Weight of Evidence: A Brief Survey. In J. M. Bernardo, M. H. DeGroot, D.V. Lindley and A. F. M. Smith (eds.), *Bayesian Statistic 2*. New York: North Holland, 249-269
- Good, I. J. (1985b). Sufficiency and Weight of Evidence. *JSCS* 21, 334-336
- Good I. J. (1988). The Interface Between Statistics and Philosophy of Science. *Statistical Science* 3, 386-397
- Good, I. J. (1989a). Yet Another Argument for the Explication of Weight of Evidence. *JSCS* 31, 58-59
- Good, I. J. (1989b). Weight of Evidence and a Compelling Metaprinciple. *JSCS* 31, 121-123
- Good, I. J. (1989c). The Theorem of Corroboration and Undermining, and Popper's Demarcation Rule. *JSCS* 31, 119-120
- Good, I. J. (1991). Weight of Evidence and the Bayesian Likelihood Ratio. In C. G. G. Aitken and D. A. Stoney (eds.), *The Use of Statistics in Forensic Science*. Boca Raton, Florida: CRC Press
- Good, I. J. (1997). Bayes Factors, Weights of Evidence and the Law. Review of J. B. Kadane and D. A. Schum, *A Probabilistic Analysis of the Sacco and Vanzetti Evidence*, New York: Wiley, 1966. *JSPI* 64, 171-191
- Good, I. J. and A. F. McMichael (1984). A Pragmatic Modification of Explicativity for The Acceptance of Hypotheses. *Philosophy of Science* 51, 120-127
- Hesse, M. (1964). Analogy and Confirmation Theory. *Philosophy of Science* 31, 319-327
- Hildebrand, D.K., Laing, J.D., Rosenthal, H. (1976). *Prediction Analysis of Cross Classification*. New York: Wiley
- Jeffreys H. (1931). *Scientific Inference*. Cambridge: Cambridge University Press
- Jeffreys H. (1939). *Theory of Probability*. Oxford: University Press.
- Kuipers, T. A. F. (1978). *Studies in Inductive Probability and Rational Expectation*. Reidel: Dordrecht
- Kuipers, T. (1984a). Two Types of Inductive Analogy by Similarity', *Erkenntnis* 21, 63-87
- Kuipers, T. (1984b). Inductive Analogy in Carnapian Spirit. In P. Asquith and P. Kitcher (eds.), *PSA 1984, Vol. I*. East Lansing, Michigan.: Philosophy of Science Association, pp. 157-167
- Kuipers, T. A. F. (1988). Inductive Analogy by Similarity and Proximity. In D. H. Helman (ed.), *Analogical Reasoning*. Dordrecht: Kluwer, 299-313
- Levi, I. (1967). *Gambling with Truth*. New York: Alfred A. Knopf
- Levi, I. (1980). *The Enterprise of Knowledge: An Essay on Knowledge, Credal Probability, and Chance*. Cambridge, Mass.: The MIT Press
- Lindley, D. V. (1964). The Bayesian Analysis of Contingency Tables. *Annals of Mathematical Statistics* 34, 1622-1643
- Maher, P. (1993). *Betting on Theories*. Cambridge: Cambridge U. P.
- Maher, P. (2000). Probabilities for Two Properties. *Erkenntnis* 52, 6-91
- Maher, P. (2001). Probabilities for Multiple Properties: The Models of Hesse and Carnap and Kemeny. *Erkenntnis* 55, 183-216
- Jerzy Neyman,

- Niiniluoto, I. (1980). Analogy, Transitivity, and the Confirmation of Theories. In L. J. Cohen and M. Hesse (eds.), *Applications of Inductive Logic*. Oxford: Oxford University Press, pp. 218-234
- Niiniluoto, I. (1981). Analogy and Inductive Logic. *Erkenntnis* 16, 1-34
- Niiniluoto, I. (1988). Analogy and Similarity in Scientific Reasoning. In D. H. Helman (ed.), *Analogical Reasoning*. Dordrecht: Kluwer, pp. 271-298
- Niiniluoto I. (1982a). What Shall We Do with Verisimilitude?. *Philosophy of Science* 49, 181-197
- Niiniluoto I. (1982b). Truthlikeness for Quantitative Statements. In P.D. Asquith and T. Nickles (eds.), *PSA 1982, Vol. 1*. East Lansing, Michigan.: Philosophy of Science Association, pp. 208-216
- Niiniluoto, I. (1986). Truthlikeness and Bayesian Estimation. *Synthese* 67, 321-346
- Niiniluoto, I. (1987). *Truthlikeness*. Dordrecht: Reidel
- Niiniluoto, I. (1998). Verisimilitude: The Third Period. *BJPS* 49, 11-29
- Egon S. Pearson
- Popper, K. R. (1963). *Conjectures and Refutations*. London: Routledge and Kegan Paul
- Romeyn, J. W. (2005). *Bayesian Inductive Logic*. PhD dissertation, University of Groningen
- Romeyn, J. W. (2006) Analogical Predictions For Explicit Similarity. *Erkenntnis* 64, 253-280
- Savage L. J. (1954). *The Foundations of Statistics*. New York: Wiley
- Skyrms, B., (1991). Carnapian Inductive Logic for Markov Chains. *Erkenntnis* 35, 439-460
- Skyrms, B. (1993a). Analogy by Similarity in HyperCarnapian Inductive Logic. In J. Earman (ed.), *Philosophical Problems of the Internal and External Worlds. Essays in the Philosophy of Adolf Grünbaum*. Pittsburgh: University of Pittsburgh Press, pp. 273-282
- Skyrms, B. (1993b). Carnapian Inductive Logic for a Value Continuum. In H. Wettstein (ed.), *The Philosophy of Science. Midwest Studies in Philosophy vol. 18*, South Bend, Indiana: University of Notre Dame Press, pp. 78-89
- Skyrms, B. (1996). Carnapian Inductive Logic and Bayesian Statistics. *Statistics, Probability and Game, Theory IMS Lecture Notes - Monograph Series Volume 30*, 321-336